Zhezhi LEI

Personal Homepage: https://cell-core.github.io/

Robot Learning | **Robot Control** | **Multi-Agent Systems**

RESEARCH INTERESTS

My research focuses on **robot learning in multi-robot systems**, with an emphasis on **efficient and safe multi-agent coordination** in complex environments. My previous work studies how multiple robots collaboratively make decisions under partial observability and dynamic conditions, integrating perception, communication, and planning. My future research aims to address key challenges in **large-scale robot collaboration**, including computational efficiency, coordination efficiency, and safety, as well as **learning and generalization in dynamic and evolving environments**.

For more details, please visit my personal homepage at https://cell-core.github.io/

EDUCATION

National University of Singapore — Singapore
M.S. in Computer Engineering | *GPA: 4.55/5* — *Aug 2023 – Jan 2025*

Harbin Engineering University — P.R.China
B.E. in Robotic Engineering, Automation | *GPA: 89/100* | *Ranking: 2/64* — *Sep 2019 – Jun 2023*

PUBLICATIONS

[1] **Zhezhi Lei**, Zhihai Bi, Wenxin Wang, and Jun Ma, "Dual-Quadruped Collaborative Transportation in Narrow Environments via Safe Reinforcement Learning," arXiv preprint arXiv:2602.16353, 2026. (Under review)

[2] **Zhezhi Lei**, Wenxin Wang, Zicheng Zhu, Jun Ma, Shuzhi Sam Ge. *Safe Motion Planning for Multi-Vehicle Autonomous Driving in Uncertain Environment*. **IEEE Robotics and Automation Letters**, vol. 10, pp. 2199–2206, 2025.

[3] Songting Liu, **Zhezhi Lei**, Haiyue Zhu, Jun Ma, and Zhiping Lin, "SegmentAnything-Based Approach to Scene Understanding and Grasp Generation," International Conference on Social Robotics, Singapore, Sep. 2024, pp. 24–30.

RESEARCH EXPERIENCE

Hong Kong University of Science and Technology (Guangzhou) — P.R.China
Robot Motion Planning and Control Lab | *Research Assistant* — *Jan 2025 – Dec 2025*

Research Topic: Dual-Quadruped Collaborative Transportation in Narrow Environments
Multi-Robot Systems | *Legged Robots* | *Reinforcement Learning*

Contributions:

- Addressed efficient multi-robot transportation in narrow environments, where coordination and collision avoidance are challenging.
- Improved training stability under shared safety constraints via advantage decomposition, mitigating non-stationarity in MARL.
- Proposed constraint allocation to enabled adaptive coordination, avoiding rigid role assignment and enhancing task efficiency.
- Validated the approach through simulation and real-world experiments, demonstrating superior performance over existing methods.

National University of Singapore — Singapore
Control and Simulation Lab | *Master Student* — *Aug 2023 – Jan 2025*

Research Topic: Safe Motion Planning for Multi-Vehicle Autonomous Driving
Autonomous Vehicle Navigation | *Collision Avoidance* | *Planning under Uncertainty*

Contributions:

- Addressed safe multi-vehicle motion planning under uncertainty, tackling noise-induced collision risks and nonlinear dynamics.
- Proposed an ADMM-based iLQG framework to handle nonlinear dynamics while improving computational efficiency.
- Developed a linearized chance constraint method to efficiently solve non-convex probabilistic collision constraints.
- Achieved higher safety with lower computational cost, outperforming existing methods especially in large-scale scenarios.

Agency for Science, Technology and Research (A*STAR) Singapore
Singapore Institute of Manufacturing Technology | Student Visitor *Jan 2024 – Dec 2024*

Research Topic: Unified Scene Understanding and Grasp Planning for Intelligent Robotic Systems
Autonomous Robot Grasping | Grasp Detection

Contributions:

- Developed a unified model for joint object segmentation and robotic grasp detection based on SegmentAnything.
- Extended a foundation vision model with a Transformer-based grasp decoder for multi-object grasp generation.
- Introduced a single-stage framework that eliminates the need for separate object detection and grasp planning modules.
- Demonstrated strong performance on benchmark datasets (OCID-Grasp, Jacquard), achieving near state-of-the-art results.

COMPETITIONS

The 21st China University Robot Competition (ROBOCON), First Prize (5th) Sep 2021 – Jul 2022
Robot Control | Computer Vision | Leader of Visual Group *C++, OpenCV; PCL*

- Designed two semi-autonomous robots for launching, grasping, and palletizing; led vision-based target recognition and deployment.
- Developed SVM-based target classification using OpenCV/PCL feature extraction, achieving 95% accuracy.
- Accelerated image processing via OpenMP-based parallelization for real-time performance.

The 11th National Ocean Vehicle Design and Production Competition, Second Prize Apr 2022 – Jul 2022
Robot Control | Computer Vision | Core Member *C, Python*

- Designed and built an unmanned rescue vehicle with vector propulsion for autonomous target localization and navigation.
- Developed a multi-steering cooperative control algorithm to enhance maneuverability on water.
- Implemented YOLOv5-based drowning detection (90% accuracy) and automated dataset generation via video processing scripts.

The 20th China University Robot Competition(ROBOCON), Second Prize Sep 2020 – Jul 2021
Robot Control | Computer Vision | Main Member *C, C++, OpenCV*

- Developed two semi-autonomous mobile robots for competition tasks.
- Designed a set of multi-target recognition algorithms with a recognition rate reaching 93%.
- Implemented omnidirectional chassis control for autonomous navigation along planned trajectories.

HONORS

National-Level Awards in China

- **Best Design Award**: The 21st China University Robot Competition — **Sole Recipient**
- **1st Prize (Rank 5th)**: The 21st China University Robot Competition — **Leader of Vision Group**
- **2nd Prize**: The 20th China University Robot Competition — **Core Member**
- **2nd Prize**: The 11th National Ocean Vehicle Design and Production Competition — **Team Leader**

Provincial and University-Level Awards

- **Third Prize in Heilongjiang Division**: The 2021 Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) — **Core member**
- **Three-Time Scholarship Recipient**: Harbin Engineering University Excellent Student Scholarship (2020, 2021, 2022) – **Ranked first in major** in 2020 and 2021.

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB

AI related Libraries & Tools: PyTorch, NumPy, Pandas, Scikit-learn, OpenCV

Robotics related Libraries & Tools: ROS, Carla, Robosuite, NVIDIA Isaac, Keil MDK